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**THE EFFECTIVENESS OF INFOGRAPHICS LAYOUT IN INFORMATION
SEARCHING AND SCANNING: AN EYE TRACKING STUDY**

A Thesis in

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by

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ABSTRACT

The layout of information arrangement could affect the effectiveness of information searching and scanning. This study aims to address the question that during information searching, how the different types of layout of infographics (linear and radial) influence readers' information acquisition behavior on first and second level of information perception and interpretation, respectively. Forty participant were recruited and eye tracker was used to record their eye movement during information acquisition. The results indicates that on the first level of information perception and interpretation, the searching performance of infographics with linear layout is as effective as infographics of radial layout; on the second level of information perception and interpretation where it requires comparison of values, the infographics with radial layout is more effective than infographics with linear layout, in terms of following metrics: time spent fixating the last of multiple targets ($t = 2.52, p = 0.018$), total duration of fixations ($t = 3.57, p = 0.001$), mean fixation duration ($t = 4.85, p < 0.001$), number of saccade ($t = 2.27, p = 0.030$), mean saccade duration ($t = -2.63, p = 0.013$), fixations/saccades ratio ($t = 4.02, p < 0.001$). Therefore, in the design of infographics, when the context of the information involves values comparison, it is recommended to arrange the information in a radial way.

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Chapter 1

INTRODUCTION

1.1 Background Information

Information is knowledge that is gained from the real world practice and is used to more accurately describe and even study the world and society we live in. As Bates (Bates, 2006) summarized, there are several types of concepts of information – natural information, represented information, encoded information and embodied information. They help people increase their understanding about matters that they don't know and decrease uncertainty when making decisions. And a variety of forms, such as text, audio, video are used as channels to convey information to the readers. Among all, text is the most widely used representation of information, in newspaper, books, etc. However, in such an information explosion era, people are so overwhelmed with the large amount of information and it becomes more difficult to manage all the information.

The development of data visualization offered another way for people to acquire information. Instead of using long and tedious words that describes complex contexts, data visualization helps people manage information more quickly and efficiently in graphical representation.

Charles Joseph Minard, a French civil engineer, was renowned for his invention in the field of information graphic. Napoleon's March (Figure 1-1) is one of Minard's

amazing work, a flow map which describes Napoleon's loss in his march to conquer Russia. In just one single flow map, many useful information that led to the defeat of Napoleon's army are conveyed, such as the geographical co-ordinates, the direction they were travelling, both in advance and retreat, the weather temperature along the paths, etc. In all, the map clearly illustrate complex information in simple graphic which is easy to read, understand and interpret.

Another pioneer who excels in visualizing data to tell a story is Edward Tufte.

Tufte insists that each data point on the graphics should have a value on its own and yet when those data points are combined for a broader inspection, trends and pattern can be observed. This is what he called as 'data-rich illustration'.

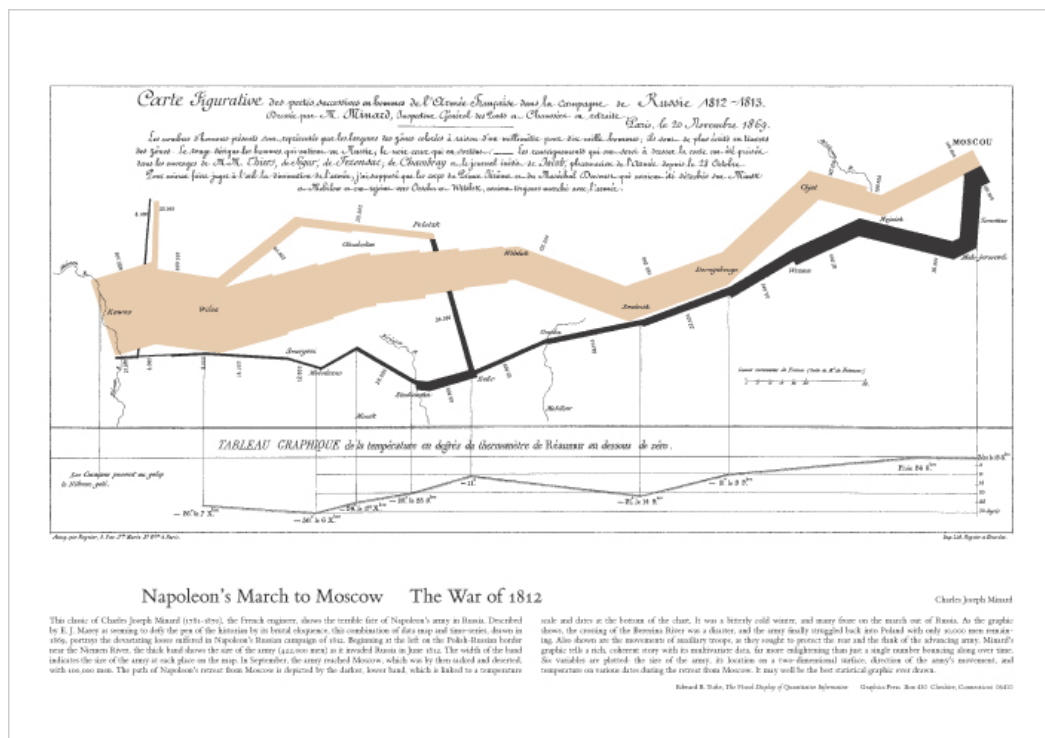


Figure 1-1 Napoleon's March by Minard

Among all the data visualization branches, information graphics or infographics are the one that are widely use in daily life such as in newspapers, modern maps, etc. The so called infographics are “*graphic visual representation of information, data or knowledge intended to present complex information quickly and clearly*” (Newsom & Haynes, 2007; Smiciklas, 2012). They are used to convey all kinds of information, news, current events, weather, and graphs for statistical data. And they could help human to see patterns and trends.

Since information graphics can express “higher lever” information as well as “lower information” (Shimojima & Katagiri, 2008), effective graphics are essential for understanding complex information and completing tasks (J. H. Goldberg & Helfman, 2010). Many research efforts have been conducted to study the factors that could make graphic more effective in different fields. Layout of the graphics is one of the hottest topic among all the other factors, such as color, font size and information forms. In the field of TV news, Josephson & Holmes (Josephson & Holmes, 2006) attempted to measure attention in different area of TV news (crawler, headline, title, etc.); (Matsukawa, Miyata, & Ueda, 2009) studied the information redundancy effect on watching TV news; and An algorithm was developed from the study of analyzing television news graphical layout (Rodrigues, Veloso, & Mealha, 2012). Study also has been done to explore the audience’s attention preferences during web navigating in terms of the positions of information and the layout composition (Chen, Yeh, Kao, Hwang, & Huang, 2012)

Study indicates that during human's cognitive processing, more than 80% of the information is acquired by eyes (Chen, Yeh, Kao, Hwang, & Huang, 2012), therefore, it is always researchers' first choice to analyze eye movement in order to evaluate the effectiveness of information graphics. For information visualization researchers, eye tracking has been a useful tool to investigate research participants' underlying cognitive processes by tracking their eye movements while they interact with visual techniques (Kim, Dong, Xian, Upatising, & Yi, 2012). Studies have been conducted to understand eye movement pattern. And many factors that influence eye movement have been confirmed. Typically, there are two types of visual behavior, visual browsing without a target and visual search with a target (M. Li, Song, Lu, & Zhong, 2009). That indicates people's visual behavior largely depends on what their purposes are while reading. Information seekers have different levels of information needs and tend to take action differently in different stages (Cole et al., 2011). This is defined as one aspect of user context, and a factor that influences visual behavior. Color, font sizes, information forms also have impacts on visual behavior (M. Li et al., 2009). Besides, domain knowledge or expertise has been shown to have significant effects on task performance and search behavior (Cole et al., 2011). Others, such as complexity, difficulty, stages, relevance judgment and usefulness have been studied excessively (White & Kelly, 2006). And another study also found that eye movement differs somewhat when reading silently and reading aloud (Fahey, Gedeon, & Zhu, 2011).

This research will look at two different layouts of information graphics, linear and radial, to see if they have impact on people's information searching and scanning

behavior and if there any patterns show up. A lot of studies have addressed the information searching and scanning pattern while reading pure text and yet there are few that look into the topic of graphics, especially information graphics. This research will take a further step and look into the field of graphics and see how layout influences people's information searching and scanning behavior.

1.2 Organization of This Research

Chapter 2 presents a review of relevant literature related to people's information behavior, layout factor in evaluating the effectiveness of the visualization, eye movement in reading, the capability of eye tracker and the metrics used in eye tracking study.

Chapter 3 will summarize the problem statement and objective of the experiment, the design of the experiment, the task and procedure of the experiment, the materials and apparatus used, and the hypothesis of the study.

Chapter 4 will present how the data were cleaned and metrics were extracted; results will be provided as well.

Chapter 5 will discuss the main findings of the study and how the study relates to the previous ones.

Chapter 6 will present conclusions of the study and discuss future work.

Chapter 2

LITERATURE REVIEW

2.1 Information behavior

Information existed in every corner of our life and is revealed to us either deliberately or unintentionally. Its definition has been discussed much in the past literature but is generally accepted as the matter that reduces the uncertainty of the unknown aspect of the mysterious world. Bates redefined the definition of information as *“the patterns of organization of everything in the universe involve every physical, biological, perceptual, and cognitive pattern of organization that exists or is extracted by sensing beings”* (Bates, 2006). He further drilled down to make clear distinctions between different concepts of information that he had developed. He stated that fundamental forms of information include natural information, which refers to all information; represented information, which is natural information after being encoded and embodied. While in daily life, we are exposed to all these forms of information through various media, visual, audial or tactile, etc. And all these channels have different efficiency in successfully conveying information and have always been a hot topic that draws researchers' attention to study how human perceive the world. There are researches that studied people's reading behavior on text (Rayner, 2001); people's ability in interpreting risks with graphs (Woller-Carter, Okan, Cokely, & Garcia-Retamero, 2012); the impacts

of text combined with graphs on reading efficiency, mental workload(Zhong, Li, Wu, & Lu, 2011), or using video and static picture to aid learning(Arguel & Jamet, 2009).

These research are often categorized as studies of information behavior, a term which was defined as “*the totality of human behavior in relation to sources and channels of information, including both active and passive information seeking, and information use*”(Wilson, 2000). And information behavior can be broken down into three sub-categories: information seeking behavior and information searching behavior and information use behavior(Wilson, 2000). Information seeking behavior is knowingly seeking behavior that helps meet a certain goal(Wilson, 2000). Information searching behavior is searcher’s behavior to acquire information through interacting with information systems on the micro-level(Wilson, 2000). And information use behavior is activities where people incorporate information into their existing knowledge systems(Wilson, 2000).

Apparently, a lot of factors could affect one’s behavior. It has been confirmed that the context of user and of task, have impact on people’s information-seeking behavior(Cole et al., 2011). From the user context perspective, when people seek for information, their cognitive ability, judgments, domain knowledge, prior knowledge, and even their needs during searching greatly affect their actions and strategies they take (Cole et al., 2011). Besides, when it comes to understanding graphical presented information, it has been proven that people’s numerical skill and graph literacy influence their performance(Galesic & Garcia-Retamero, 2011); individual who has poorer numerical skill and graphical literacy appeared to have difficulty in comprehending

information on graphs(Galesic & Garcia-Retamero, 2011). On the other hand, task context changes people's information-seeking behavior as well. A facet classification of task has been established that summarized and integrated the results of task classification in past researches. The facets of tasks that have impacts on information-seeking behavior include the source of task, whether it is internal motivated by a task doer or is external appointed by task setters; the familiarity of the task, whether it is a special task that task doer never encounter before, a more familiar task that a task doer does not conducted much often or a routine task; the goal of the task, whether it is specific goals, abstract goals, mix goals, single goals and multiple goals, etc., (Y. Li & Belkin, 2008). The visual search behavior could also be affected by the representation of the information, such as layout of the information, and such as its colors, font size, etc(M. Li et al., 2009).

In previous research, it has been pointed out that information graphics consist of a minimum of three different stages in information perception and interpretation(J. Goldberg & Helfman, 2011). The first level of information perception and interpretation is the lookup of values. A simple task would be to find out how much the rainfalls is in millimeters on average in July in the bar graph given. The second level of information perception and interpretation is the comparison of values. An example task would be to find out which month has more rainfalls on average in the bar plot give, July or August. The third level of information perception and interpretation is the comprehension of the trends and distribution, etc. An example task would be to describe the trend of rainfalls in milliseconds from January to December on average. Past research have been focus on the basic level of the information perception and interpretation(J. Goldberg & Helfman,

2011) and/or the second level of the information perception and interpretation(J. H. Goldberg & Helfman, 2010).

The following are some of the metrics that were reported in previous studies(J. Goldberg & Helfman, 2011; J. H. Goldberg & Helfman, 2010) by research to measure the efficiency of information seeking tasks.

Response time (seconds): After participants confirmed that they've already understood the questions/task requirement, with the appearance of a graph, the starting point of response time is recorded; when participants respond with their verbal answers, the ending point of response time is recorded.

Minimum time (seconds): the theoretical minimum time for a participant to locate the correct piece(s) of information by going through area(s) of interests that he is supposed to look at.

2.2 Layout factor in visualization evaluation

With the exponential increase of the information in this era, studies of various kinds have been conducted to evaluate the effectiveness of the representation of information in visualization whether this information is stored in physical objects like books, magazines, etc., or is showed digitally in different kinds of file on displays and monitors.

As mentioned in previous section that there might be various factors, such as user/task context or font, color and layout, it is found that many studies has been focused on examining the relationship between the layout of information and the effectiveness in information perception and acquisition.

Zaphiris & Kurniawan (2001) conducted a research to investigate people's reading speed between paper and monitor presentation. They specifically chose the factor of information layout to dig deeper in this topic, among all the other known factors such as angle of the reading material, lines per page, visual angel of the character, resolution of the displays, familiarity with medium, etc. In their study, the layouts of information that were tested are one-column, two-column, and three-column reading material obtained from a sample ETS test (Zaphiris & Kurniawan, 2001). The results from this study showed that neither there was significant difference between numbers of column and the reading speed, nor there was difference in user's preference towards the numbers of columns in reading (Zaphiris & Kurniawan, 2001).

Dyson (2004) also conducted a more comprehensive study to investigate the effects of text layout on people's reading from screen. The variables that were looked into with respect to text layout were line length(characters density, type size, physical length, characters/line, characters per line and margin), columns (one- vs. two- vs. three-columns), window size(different lines per screen) and interlinear spacing(single vs. double spacing), with a strong focus on line length based on the large amount of previous studies(Dyson, 2004). The results from this study were that numbers of character per line is a significant variable that affect reading and suggested that while investigating one

variables, one should keep the rest of variable consistent so that no confounding variable would be introduced and damage the validity of the results(Dyson, 2004).

Gradually, researchers started to shift their attention to the information arrangement on the web from an academic perspective and tried to study the layout of web pages. Li et al.(2009) divided the webpages into four sections, quadrant I (upper right), quadrant II (upper left), quadrant III (lower left) and quadrant IV (lower right), with an additional section in the center area added, making each section take up around 20% of the page in area (M. Li et al., 2009). Ten web pages of the same context were designed except for one difference: a piece of target information was rendered in two forms, text or picture, and was placed in the one of the five sections at a time. Participants were required to locate and click on the correct target and searching efficiency was collected(M. Li et al., 2009). The finding was that it takes shortest time to successfully target the intended information in quadrant II (upper left) for both forms of text and picture and people has worst performance, meaning longest time, when placing text in the quadrant IV (lower right) and picture in quadrant III (lower left)(M. Li et al., 2009).

Another group of researchers also investigated the positions of information and the layout composition of web pages based on audience's attention preference (Chen et al., 2012), but they combined text with picture and control the relevant position of text to photos; three types of layout composition were devised with one being photo on the left and text on the right, another being photo on the write and text on the left, and the last being photo on the top and text on the bottom(Chen et al., 2012). The materials that were used to test the participants in this study were the news that were tailored according to the

three layout composition described above. Another factor that was taken into consideration was where to put the news in the web pages(Chen et al., 2012). As a result, the web pages were divided into three blocks equally, with one on the left, one middle, and one right. Three pieces of information (news) with different layout composition were randomly placed into the three blocks and participants were asked to navigate through all the presented information(Chen et al., 2012). The results indicates that there was no significant difference among the layout composition, but the block on the left did attract the most attention of the participants(Chen et al., 2012).

Besides choosing text as the research material, Goldberg and Helfman decided to step further to study the layout of graphical presented information – information graphics which is not the type of information graphics or infographics that was mentioned in Chapter 1. Particularly, linear and radial graphs were studied where linear graphs are graphs, such as bar chart, line chart, etc., that organize value along the perpendicular dimensions and while radial graphs, such as radar graphs, arrange values along the axes that spread out from the center(J. Goldberg & Helfman, 2011). It is claimed radar graphs are more effective than bar chart in assessing the symmetry of values and in locating extreme values, and that bar charts are more useful if one need to compare their magnitude(J. Goldberg & Helfman, 2011). In the study, participants were asked to in first type of task look up a value in single dimension and in second type of task to add or subtract values from two dimensions(J. Goldberg & Helfman, 2011). The lookup of a value in the first type of task is the level one of information perception and interpretation; Adding or subtracting values on two dimensions didn't fall into any one of the

information perception and interpretation levels, but seems to be the combination of two level-one perception plus a simple mathematical calculation. As for single-dimension lookup task, better performance was obtained from linear graphs than radial graphs with respect to response time and scanpath-computed minimum times and it was found and verified that participants followed a three-stage information searching process: (1) find dimension, (2) find data point, (3) get value. As for two-dimension value lookup task, participants were found to spend more time completing the task with radial graphs and linear graphics(J. Goldberg & Helfman, 2011). In the discussion, Goldberg and Helfman did admit that the design of the study focused more on the controlled experiment rather than on making sure the tasks were appropriate for the graphs type(J. Goldberg & Helfman, 2011). In the end, guidelines were put forwarded to lay a scientific foundation for design as reference.

Goldberg and Helfman also carried out another study where tasks type were designed based on the second level of information perception and interpretation(J. H. Goldberg & Helfman, 2010). Bar graphs, line graphs and spider graphs were still used as study materials, and tasks have different difficulty level (hard and easy) on depending on the number of dimensions and targets were used to complete the task. Easy task were designed such that participants were required to use two dimensions to make a decision on the target, whereas hard task required to choose between two targets across eight dimensions. In terms of completion time, graph type and difficulty condition were not significant; in terms of scanning in hard tasks, the first time when participants viewed bar

graphs, a left-to-right strategy was not followed; spider graphs generated much more back and forth cross-scanning between the two overlaid spider graphs.

Though the studies mentioned above all explored the effects of different layout of the information, they were somewhat different with respect to the definition of layout. Since both of Zaphiris & Kurniawan's and Dyson's studies focused on text layout, they have a similar definition of information layout which is the physical arrangement of the whole contents(text), whether it is one or two columns, ten or twenty characters per line, etc., except that Zaphiris & Kurniawan's was much simpler. Besides, Li et al. studied that which forms of information, text or picture, was easier to locate. Li et al.'s was different in the sense that only the change of the location of one piece of information (five section), not the arrangement of whole information, was considered in a global environment (web pages). Chen et al.'s was again different the previous three. In their study, not only the location of the pieces of information (left, middle and right) were changed, but also the relative position of text and photos within the piece of the information was also considered a variable in the experiment. Regarding both of Goldberg and Helfman's experiments, the layout factor was completely a different explanation; it is the graphical layout of graphs themselves and it has more to do with the arrangement of the value along with the graphs that helps further elaborate the values.

2.3 Eye Movement

Information processing during visual perception is strongly influenced by constraints arising from the human oculomotor system (Engbert, Kliegl, & Longtin, 2004). And also because of acuity limitation in the visual system, eye movements are needed during reading (Rayner, 2001).

One of a common misunderstandings about eye movement is that most people think that eyes move smoothly across text when they are reading. However, eye movement is made of series of movements that are called saccades during which no information is acquired because eyes move very fast, and between saccades there are fixations during which eyes are relatively still and this is the stage that give eyes enough time to comprehend the text (Martinez-Conde, Macknik, & Hubel, 2000). Another misunderstanding about eye movement is that theoretically, the way people read written language is normally considered from left to right. In fact, the actual reading were found involved events such as multiple fixations on a word (refixations), skippings of words and regressions to previous words (Engbert et al., 2004).

Eye movements in reading are affected by many factors; personal behavior and previous experience are two of those factors. Since child, we are taught to read from left to right and as we become more and more skilled in reading, we developed our own reading behavior without even noticing it (Vo, Mendis, & Gedeon, 2010). Some prefer to scan and skim while other read word by word. Study also showed that eye movements are different when reading silently and reading aloud: people spend more time processing the

information when reading aloud or while listening to a voice reading the same text than in silent reading (Fahey et al., 2011). Also, when reading passages of text that are hard or unfamiliar, skilled readers often have to take different actions and strategies to adaptive the difficulty level of the text (Rayner, 2001). They spend more time reading and less time transitioning between pieces of information and tend to look back and forth repeatedly (Rayner, 2001).

On the other hand, the text itself and its composition also greatly influence one's visual behavior. Extensive researches have been done to examine the variable that might affect reading from screen. The text layout variables is one of them and includes line length, columns, window size and interlinear spacing, etc (Dyson, 2004). There was argument from typographers that typographical variables should not be consider alone (Dyson, 2004). For example, long line lengths require more interlinear spacing so that the eyes could transit accurately between the end of this line and the beginning of next line (Bouma, 1980). This is because the if angle of the return sweep is too small, it is difficult to accurate locate the next line (Bouma, 1980). Lund (Lund, 1999) agreed with argument that typographical variables should not be consider alone, otherwise they would confound with other variables (Dyson, 2004). For instance, line length can be measured by the physical length of the line or numbers of characters in the line, but it is also determined by the type size of each character, hence the type size and number of characters per line are confounded (Dyson, 2004). From the previous studies, conclusions can be drawn that the text layout do have a complex effects on visual reading behavior.

2.4 Eye tracking

An eye tracker is a device that can accurately detect the movement of eyes and the gazes by tracking pupils' position and corneal reflection. The first eye tracker could be traced back to late 1800s, when the eye tracker was built mostly mechanically and was very unconformable to wear (Holmqvist et al., 2011). And researchers throughout most of the twentieth century had to build the eye tracker of their own to continue with their studies, which led to various problems including hindering their research process, and making eye tracker complete impractical (Holmqvist et al., 2011). It was not until mid-1970s that engineer-driven company such as ASL (Applied Science Laboratories) were founded to build eye trackers for researchers (Holmqvist et al., 2011). However, this also brought a lot of troubles between manufacturers who didn't have enough knowledge about what eye tracker were used for in researches, and researchers, who didn't know the mechanism of the eye tracker and influence the further development of existing eye trackers (Holmqvist et al., 2011). Fortunately, things are getting better; more and more manufacturer strive to make eye tracker more easy to use and over past 20 year, the number of researchers who used eye trackers in their studies greatly increased. Nowadays, leading manufacturers of eye trackers include Applied Systems Laboratory, SR Research with EyeLink system and Tobii Technology, etc.

There are three type of video-based eye tracker including static eye tracker, an eye tracker that has little or no contact with the participant and usually is place on the table; head-mounted eye tracker, which place both illumination and cameras on the head of the participants, often frame on either a helmet or a pair of glasses, etc.; and head-mounted

eye tracker with addition head tracker to compensate the effects of head movement on the real performance(Holmqvist et al., 2011).

Several types of events are commonly used to describe eye movements: fixation, saccade, glissade, smooth pursuit, micro-saccade, tremor and drift. Eye trackers are said to track the movement of eyes, but in fact, most of the eye trackers report how long eye fixate on a certain position over a period of time(Holmqvist et al., 2011). Such event is called fixation, which is somewhat misleading, because eyes can't be complete still and have micro movements which can be classified into three sub-categories: drift, tremor and micro-saccade(Martinez-Conde et al., 2000). More specifically, drift might be result of the inaccurate control of eye muscle, but it is always slow movement that make the focus of eyes deviate from the center of fixation within a certain range; mircro-saccades are the relevant faster movement that takes focus of eyes back to where it should be during fixation; and the role of tremor is unknown so far(Martinez-Conde et al., 2000). It is generally accepted among researchers that it couldn't be called a fixation unless there the eye has fixated for at least 100ms, which is commonly used fixation threshold in eye tracking study, but it might differ depending on the nature of the task.

The quick movement of the eye is called saccade which eye moves at least at a velocity of 30 arc degree per second. Saccades are fast and short and are the fastest movement human can make. Therefore, generally, it is widely agreed that no information is acquired during saccades(Martinez-Conde et al., 2000). Glissade are the movement refinements to reach a target after saccade movements end(Martinez-Conde et al., 2000). Smooth pursuit happens when eyes slowly follow a moving target. Although saccades

and smooth pursuit are eye movements, they are driven by different part of the brain and hence not the same type of movement(Martinez-Conde et al., 2000). There are also generally accepted thresholds based on previous studies for events like saccades and glissades, which make it easier for researches to detect these event with algorithms when necessary. See Figure 2-1 for threshold values of eye tracking events.

Type	Duration (ms)	Amplitude	Velocity
Fixation	200–300	–	–
Saccade	30–80	4–20°	30–500°/s
Glissade	10–40	0.5–2°	20–140°/s
Smooth pursuit	–	–	10–30°/s
Microsaccade	10–30	10–40'	15–50°/s
Tremor	–	< 1'	20'/s (peak)
Drift	200–1000	1–60'	6–25'/s

Figure 2-1 Thresholds for eye tracking events

In the events detection of eye movements, some of the experience eye-tracking researchers prefer to subjectively set the threshold, for example velocity threshold, based on their past experience and the task type. Or many other researchers tend to make use of existing algorithms to detect events and analyze eye movement.

There are five basic algorithms(Salvucci & Goldberg, 2000) and even more developed and adaptive algorithm(Nyström & Holmqvist, 2010), but two most popular algorithms are dispersion-based algorithm and velocity-threshold algorithms. Velocity-threshold algorithm is simple and easy to understand; if the velocity of a data point recorded is greater that the velocity threshold, then this data point is classified as saccade. Any data points that are not saccades are fixations. And group consecutive fixations data

points and calculate the centroid of all these point and used as the center of this large fixation (Salvucci & Goldberg, 2000). The limitations of this algorithm are (1) the use of velocity-threshold algorithm requires high sample frequency (more than 200 Hz), otherwise there would be large errors(Martinez-Conde et al., 2000); (2) though there is a universally accepted range for saccade velocity, the setting of velocity threshold is too subjective; different researcher chose different velocity and the varies a lot (3) the threshold varies from task to task; if the threshold is too small, then a lot of small saccades are included; if it's too large, then a lot of data points are grouped as a rather large fixation which shouldn't exist at all; (4) as the unit of velocity in literatures is arc degree/ second, the transformation of actual eye moving distance to eye movement in arc degree needs to be obtained in order to compare the actual velocity with the threshold. If can't, then it needs some exploratory data analysis to empirically determine the threshold(Salvucci & Goldberg, 2000).

Dispersion-based algorithm, on the other hand, is more complicated than velocity-threshold algorithm. It makes use of the fact that fixation data points are normally gathered together because of low velocity. A duration threshold for a fixation is one of the parameters in this algorithms, which usually would be set as 100 ms. So the typical way of doing a dispersion-based algorithm is to generate a time window of 100 ms and move this time window to collect the data points and the dispersion of all the data points within it to see if it exceed the dispersion threshold; If not, then add another data point to this group and see if the dispersion is larger than the threshold; continue doing so until it exceeds the threshold, then the group of data, excluding the new added one, are collapsed

as a fixation; use the centroid of the data points as the position of this fixation; repeat the same with a same time window starting where the last data point is excluded until all the data points have been through (Salvucci & Goldberg, 2000). The advantages of this algorithm are that it is more accurate to identify fixations because of its clustering mechanism behind it. However, since there are two parameters for researchers to determine – dispersion threshold and duration threshold, in reality it might not work very well because of the conflict between the two parameters and the more the parameters one has to set, the more likely it might introduce unexpected errors.

So which algorithms to choose? There is always hot debate on between “saccade-picker” and “fixation-picker” (Karn, 2000), but the recommendation is if the sampling frequency is low, dispersion-algorithms has better performance and otherwise, velocity-threshold is worth trying (Martinez-Conde et al., 2000).

In the evaluation of eye tracking experiment, Jacob & Karn summarized the metrics used in 21 usability studies incorporating eye-tracking, most commonly used metrics were reported (Jacob & Karn, 2003). These include (Jacob & Karn, 2003):

Number of fixation overall: the number of fixation overall during search behavior; the large the number, the less efficient the search; researchers should consider the time effects on the number of fixations, the longer the time, the more fixations in number.

Gaze % (proportion of time) on each area of interest: the percentage of time spent looking at a particular area of interest.

Fixation duration mean overall: average time spent on each fixation point; the longer the fixation duration mean, the more difficulty for participant to extract the information from the material.

Number of fixations on each area of interest: self-explanatory; the large number of fixation on a certain area of interest, the more important the target is.

Gaze duration mean on each area of interest: it is similar to the fixation duration mean but more narrow down to each area of interest; the larger the mean, the more difficult that area is to understand.

Fixation rate overall: number of fixation per second.

There are also some other promising metrics that were used in different studies, including (Poole, 2005):

Number of saccades: saccades can also be defined as the movement between two fixations, therefore number of saccades equal to (the number of fixation – 1); the larger the number, the less efficient the search (same with the interpretation of number of fixations) (J. H. Goldberg & Kotval, 1999).

Saccade amplitude: eye movement in arc degree; the larger the amplitude, the longer distance eyes need to travel and such indicate the more effective search of useful information clues (J. H. Goldberg, Stimson, Lewenstein, Scott, & Wichansky, 2002).

Scanpath duration: equals to the sum of each saccade duration; longer scan path, less efficiency the search (J. H. Goldberg & Kotval, 1999).

Scanpath length: same interpretation as scanpath duration, except in length(J. H. Goldberg et al., 2002).

Fixation/saccade ratio: the total time spend fixation vs. time spend on saccades; the larger number represent more information processing than information searching(J. H. Goldberg & Kotval, 1999).

Time to first fixation on-target: the time from beginning to the time fixation is first locate on the intended target (Byrne, Anderson, Douglass, & Matessa, 1999).

Time to first fixation using multiple targets: when multiple targets are used to complete task, the time from beginning to the time eyes fixating on the last target(J. H. Goldberg & Helfman, 2010).

Time between first target fixation and first target click: completion time minus the time to first fixation on intended target; it is a metric describe the uncertainty of a participant in the comprehension of the information(J. H. Goldberg & Helfman, 2010).

Besides the metrics above, scanpaths was also another method to analyze eye tracking data. It was generated to visualize the search and scan pattern and describe how eyes physically move. Each scan path is actually a broken connecting all the fixations points in a sequential way (time). The representation display not only the direction of eye movement, but also the time eye spends on each fixation. The larger the fixation point in circle form, the longer eye stays.

Scanpath was used a lot in the past researches(Buswell, 1935; Holšánová, 2001; Josephson & Holmes, 2006). The purposes of using scanpaths were to check data quality,

get first impression of the data, or analyze data manually (Martinez-Conde et al., 2000).

Scanpaths could also be used to access the similarity between two paths with a string-edit calculation (Foulsham & Underwood, 2008). Briefly speaking, an 5 * 5 grid was overlaid on the actual scanpaths and difference between two scanpaths was calculated by counting the editing cost when altered one path to another (Foulsham & Underwood, 2008), see Figure 2-2 String-edit method for the introduction of this method from the previous research.

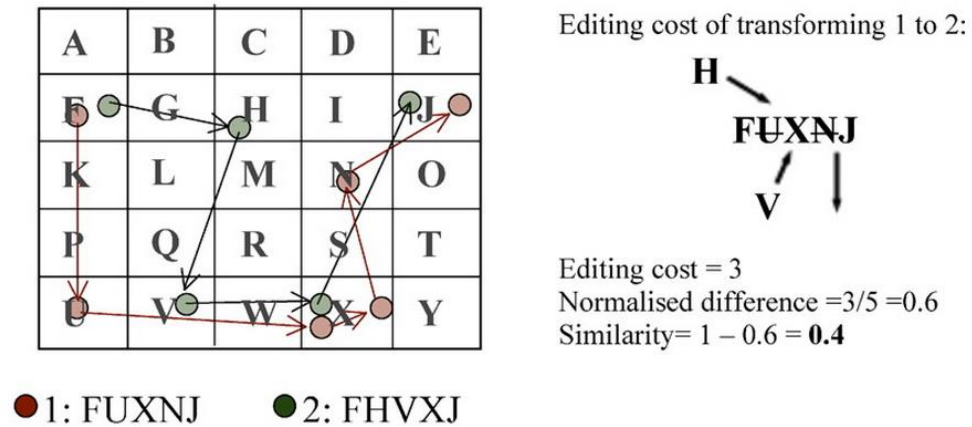


Figure 2-2 String-edit method

Chapter 3

METHODOLOGY

3.1 Problem Statement and Objectives

Infographics are graphic visual representations of information, data or knowledge intended to present complex information quickly and clearly (Smiciklas, 2012). Providing effective graphic visualization greatly determine the efficiency of conveying complex information, factors of various kinds were studied in previous research to develop guidelines for designing more effective information transmission media. Previous studies (J. Goldberg & Helfman, 2011; J. H. Goldberg & Helfman, 2010) investigated the layout factor of visualization forms (bar graphs, line graphs and spider graphs) of graphics themselves on the first two levels of information perception and interpretation. Yet research hasn't been conducted in a more complicated context with respect to the amount and representation of information that are exposed to the readers. This study was designed to use eye tracker to investigate the effectiveness of information searching behavior and scanning behavior of participants in the setting of different graphics layouts.

The two different types of layout that were examined in this study are linear layout and radial layout. Linear layout is the organization of information in a linear form, either from left to right or from top to bottom, whereas radial layout is the organization of information from center outward. The definitions of searching and scanning are different from what were defined in the previous literature (Wilson, 2000). They are generalized

based on people's information acquisition needs in daily life, whether it is seeking information with a purpose or reading selectively without any purposes (such as reading magazines, etc.)

The study mainly aims to address the question that during information searching, how do the different types of layout of infographics influence readers' information acquisition behavior on first and second level of information perception and interpretation, respectively?

Additional questions that are attempted to be addressed are:

- Do any patterns exist in information searching?
- Do any patterns showed up in information scanning?

3.2 Participants

Forty students (23 males and 17 females) were recruited from the Pennsylvania State University in this study. All the participants are graduate students from different departments, such as Industrial Engineering, Kinesiology, Animal Science, etc. 32 participants are in the age range from 18 - 25, and the rest 8 participants are in the range from 26 – 40. They had no vision problems and/or wore corrective lenses.

3.3 Equipment

An Arrington Research head mounted eye tracker with scene cameras and a binocular tracking system was used in this study, running view point PC-60 software that collected the eye tracking data. The eye tracker was connected to a desktop computer with a 27 inches display (1920 × 1080 pixel). A neck pillow was used to mitigate the fatigue of participants due to the requirement of remaining head still during the experiment. A laser pen was used to pinpoint the calibration points on the wall before the experiment started. And a measuring tape was used to measure the view distance of each of the participants from the surface of the display to their eyes. In addition, a 30 Hz sampling frequency was used to collect the eye movement data. Analysis was conducted using a combination of post-hoc analysis software that came with the eye tracker, R cran, Minitab and Microsoft excel.

3.4 Graphics Stimuli

The graphics stimuli used in this study include one actual infographics and its altered-layout representation and two other actual inforgraphics selected by researches from the infographics pool on the internet.



Figure 3-1 Every Minute of the day (Radial) (left)

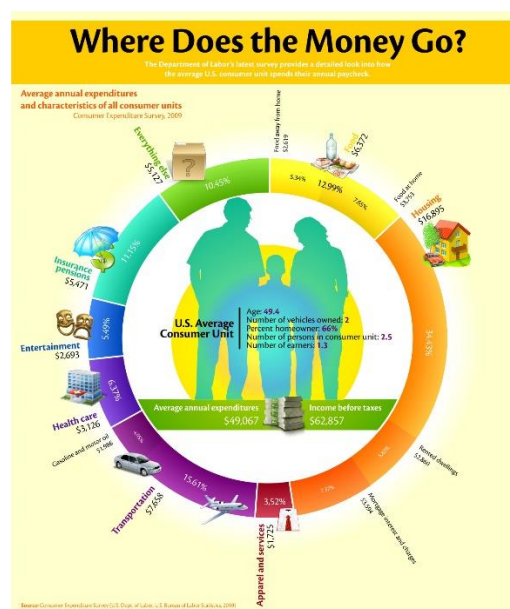


Figure 3-2 Where does the money go? (Right)

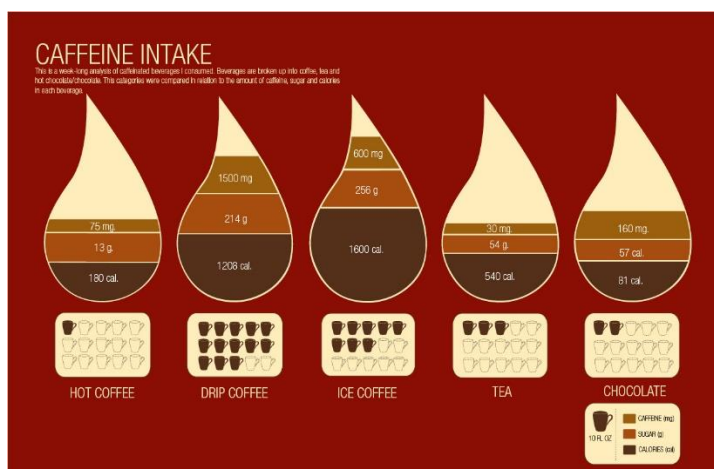


Figure 3-3 Caffeine Intake

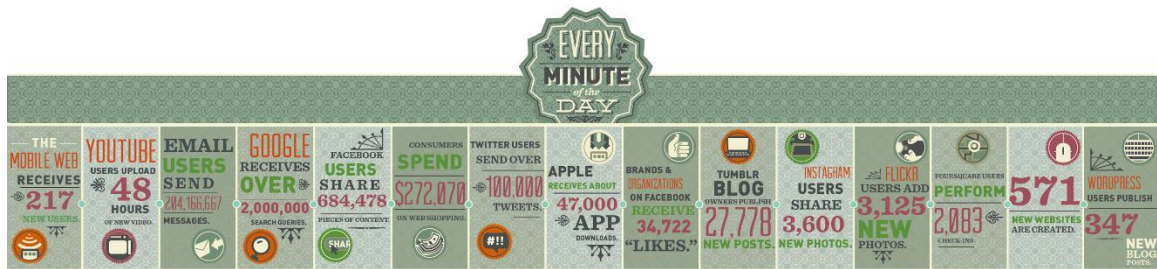


Figure 3-4 Every Minute of the day (Linear)

The contents of infographics chosen were carefully filtered in a way that the population (graduate students) is either familiar with the content or not at all. Figure 3-1 is a picture of infographics that describe what happened in every minute of the day on the internet; Figure 3-4 is its altered-layout representation of the same infographics. Figure 3-2 tells a story of how much do consumers spend their annual paycheck and Figure 3-3 describe how much does the designer of this picture intake caffeine, sugar and calories in each type of the coffee.

3.5 Procedure

Each participant was tested individually for around 25-40 minutes in the experiment. After participants sign the Informed Consent Form for the experiment, general orientation briefing was given to participants introducing the purpose of study, apparatus and the steps they needed to go through in the following parts of experiment. They were free to ask any questions about the experiment at this stage.

Then a calibration session was followed to make sure that the eye tracking device is able to successfully track the pupils of participants and hence guarantee accuracy of the

eye movements recorded. A neck pillow was placed on the participants and they were asked to adjust their sitting position and distance to the display in a way that they felt comfortable seeing the content on the display and could freely use the mouse without moving their bodies. Then participants were asked to turn to their right facing the wall where calibration happened. The distance between the wall and the participants' eye was the same distance when they sat in front of the display. A laser pen was used to pinpoint the 16 calibration points on the wall that were generated by the View Point PC-60 software randomly. During calibration, participants were required to remain their head still as much as possible and to not blink while they were looking at the exact point. They were free to blink and look around between calibration points. The calibration session ends when participants successfully targeted the 16 calibration points and they turned to their left facing the display again with the same view distance as before.

Next, a short training session was provided to ensure that participants got familiar with apparatus and task type. In the training session, participants was showed two picture of infographics picked by researchers (different from the pictures in the real experiment trials). Participants were asked to remain their head still, feel free to ask questions if there were understanding problems before they see the pictures and not to use the cursor to guide their reading. Then they performed searching task to the first picture of infographics and answered the question followed and reported their confidence in the answer they chose; and then scanned the second picture of infographics to understand the broad picture the infographics tries to convey to the readers after which an understanding question was showed and they were required reported their confidence in the answer.

After the training session, researchers repeated the purpose of each task and made sure participants fully understood the process.

Next was the real experiment trials where participants were showed four pictures of infographics. Research repeated the same requirements as in the training session before they started. The first two trials were related to searching and the rest two scanning. The first two trials showed exact the same picture of infographics (either Figure 3-1 or Figure 3-4, depending on what group they were in). The task in first trial was to find out “*How many tweets do twitter users send in every minute of the day*”. After they read the picture, they were asked to choose the answer of a multiple-choice question of the exact numbers of twitter send in every minute of the day and they reported whether they were sure about or guessed the answer. The task in the second trial was to find out “*What does the second largest number in the infographics describe*”. Questions was followed to confirm that they obtained the correct information by giving a multiple-choice question asking about the description of the second largest number in the infographics and they reported whether they were sure about or guessed the answer. There were no tasks for the third and fourth trials (scanning part). Participant read pictures of infographics based on their reading own behavior and they stopped when they thought they understood the information. In the third trial, Figure 3-2 was showed and the question for this infographics was “*What are the top three categories in which average consumer spends their annual paycheck*”. Then they reported their confidence in the next question. In the last trial, Figure 3-3 was displayed and question was “*On average, drinking a cup of ice coffee take in the most sugar, caffeine and calories; is this statement*

true from the previous infographics”. Then they reported their confidence in the answer.

And that’s the end of the experiment trials. The task order is different between groups, see Figure 3-5.

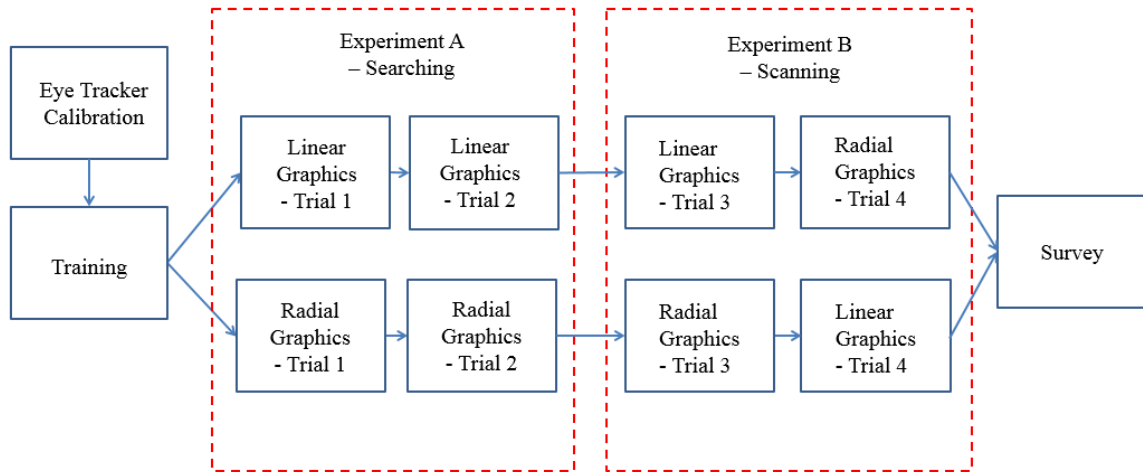


Figure 3-5 Task order and general procedure

Lastly, participants needed to choose their gender and age range in another file and the whole experiment ended.

3.6 Design of Experiment

The first two trials were designed as between subject experiment with an independent variable being the layout factors. The rest of trials were two individual experiment aiming to descriptively and qualitatively analyze people’s scanning behavior.

3.7 Variables

(1) Between subject experiments (Trial 1 and 2)

Independent variable: Linear layout and radial layout

Dependent variables:

Performance variables: completion time, accuracy score.

Eye tracking variables: time to 1st fixation on intended target, time to 1st fixation on the last of multiple targets, time after 1st fixation on intended target, time after 1st fixation on the last of multiple targets, total duration of fixations, mean fixation duration, total duration of saccades, mean saccade duration, total amplitude of scanpath, mean amplitude of saccades, fixation / saccade ratio.

(2) Individual experiment (Trial 3 and 4)

Dependent Variables:

Performance variables: completion time, accuracy score

Eye tracking variables: total duration of fixations, mean fixation duration, total duration of saccades, mean saccade duration, total amplitude of scanpath, mean amplitude of saccades, fixation / saccade ratio.

3.8 Hypothesis

Based on the findings of Goldberg and Helfman (2010), the hypothesis is that linear layout is more effective than radial layout with respect to participants' task performance in the following metrics:

- (1) Completion time
- (2) Accuracy score

Eye tracking metrics were meant to use to further elaborate and support the analysis.

Chapter 4

DATA ANALYSIS AND RESULTS

4.1 Data Cleaning and Extraction

4.1.1 Performance metrics

Performance metrics in this study include completion time and accuracy score.

Completion time was collected by timing when the trail started and when it ended.

Another performance data collected from the experiment is the accuracy of participants' answers to the questions and their confidence about their answer.

In the experiment, participants were request to answer two questions per trial: one is an understanding questions that asked them about the specific information in the previous infographics and the other is a question that measured their confidence in the answer – whether they had confidence or were just guessing. Therefore, accuracy score consists of two part: one is whether a participant chose the correct answer; the other is whether a participant was sure he chose the right one. Appendix A shows the summary of all the participants' answers where 1 in the coding represents the participant made the correct choice and 0 the wrong choice; Y stands for participants have confidence in knowing they get the right answers whereas N stands for participants were just guessing.

Score accuracy of each participants is obtained through the below scoring table created by the researchers. The rationale behind the scoring table is that if a participant answer correctly and is certain about the answer, one get the highest score; if one answer correctly but is not sure, based on the performance, one still did it well so one gets the second higher score; certainly, if one was not sure and chose the wrong one, at least one had good self-evaluation of oneself, he goes to the next ranking; if one is certain and chose the wrong one, he neither got the answer right nor correctly evaluated himself, hence the lower score. There are four different performance in total, therefore the best performance scores 4 and worst 1.

Table 4-1 Accuracy and confidence scoring table

Scores	Answer correctly (1)	Answer wrongly (0)
Certain (Y)	4	1
Uncertain (N)	3	2

4.1.2 Eye tracking metrics

The eye tracking data were recorded by View Point PC-60 software of Arrington Research at the sample frequency of 30 Hz. Binocular mode with parallax correction was used to more accurately detect the eye movement. Because of the corruption of part of the data, only data points from right eye were extracted from the raw data and used to generate metrics. However, after accurate calibration, the difference of correct X and of

Y coordinates of between two eyes are unnoticeable (<0.0001), that means the use of data points from one eye is still valid to conduct data analysis. Blinks and poor quality data points were deleted from the data sets and total time and delta time were recalculated for different task. A velocity-threshold algorithms was first used to detect fixations and saccades, but the results turned out to be very inaccurate due to the low sampling frequency. A dispersion-detection algorithms was then implemented to collect more accurate metrics from the cleaned data. The time threshold for the dispersion method is 99 ms where it was commonly set as 100 ms. The reason is that 30 Hz sampling frequency was supposed to record each data point at every 33.33ms, making 100ms include exact three data points; however, due to sampling error, in most cases the total duration of three data points are below 100ms yet above 99ms. Choosing the threshold of 100ms would introduce large error in the results after running exploratory data analysis. Therefore 99ms was chosen as the time threshold in the dispersion-detection algorithm. The dispersion threshold was set as 1 arc degree based on the literature. However, the software could only record the distance of eye movement on the display in normalized coordinates from 0 to 1. Therefore, extra step was performed to transform the normalized distance on the display to arc degree used to measure pupil movement. The relationship between the moving distance on the display and the arc degree is that the farther one sits away from the display, the longer the moving distance is per arc degree. As the view distance of each participants varies, the transformation coefficient differs person by person, but could still be extracted from the post-hoc analysis software of Arrington Researching by adjusting the units in the system.

Eye tracking metrics were then extracted from the events file generated by the algorithm which include information such as event type (fixation or saccade), starting time, lasting time, starting coordinates and ending coordinates (only exist in saccades).

4.2 Results by trial

Prior to data analysis, data transformation were done and marked with * on the metric in the following section on. Normality and equal variance assumptions have been checked, and an alpha level of 0.05 was used for all statistical tests. And independent two-sample t test was selected for parametric data and Mann Whitney U test for non-parametric data to examine the significance level of the metrics.

Two outliers in the data were identified through scanpath inspection. One was due to the pupil tracking failure during the experiment; the other was corrupted because the participant adjusted the device and pause the searching in the middle of the experiment. All the metrics of these two outliers were deleted in all of the following trials.

Table 4-1 summarizes the metrics that are significant difference in the first two trials.

Table 4-2 Summary of Metrics and their significance

	Significant
	Marginally Significant
	Not Significant
	N/A

		Trial 1	Trial 2
Performance Metrics	Completion time		
	Accuracy score		
Eye Tracking Metrics	Time to 1 st fixation on intended target		
	Time to 1 st fixation on the last of multiple targets		
	Time after 1 st fixation on intended target		
	Time after 1 st fixation on the last of multiple targets		
	Total duration of fixations		
	Mean fixation duration		
	Total duration of saccades		
	Mean saccade duration		
	Total amplitude of scanpath		
	Mean saccade amplitude		
	Fixations/Saccades ratio		

4.2.1 Trial 1

Trial 1 aimed to examine the information searching behavior difference between linear and radial layout on the first level of information perception and interpretation – value lookup.

Several independent two sample t-test were conducted to Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) on all metrics. There are no significant difference on the performance data and none of the eye tracking metrics show significant difference except for one metric - mean fixation duration which

is marginally significant, with linear layout infographics ($M = 213.1$ ms, $SD = 40.8$ ms) and radial layout Infographics ($M = 190.3$ ms, $SD = 22.7$ ms); $t(33) = 2.01$, $p = 0.052$. This might indicate that on average, participants spending more time processing the information on linear layout than radial layout.

The rest of the metrics that are not significant can be found at the Appendix B.

4.2.2 Trial 2

Trial 2 aimed to examine the information searching behavior difference between linear and radial layout on the second level of information perception and interpretation – values comparison.

(1) Performance metrics

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **completion time**. There was a significant difference in terms of completion time for linear layout ($M = 29.439$ s, $SD = 13.127$ s) and radial layout ($M = 21.947$ s, $SD = 8.936$ s); $t(33) = 2.09$, $p = 0.044$. This indicates that it is significantly faster to finish the comparison task for radial linear than linear layout.

A Mann Whitney U test to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear). The result indicated that **accuracy score** was not significant between linear layout ($Mdn = 2.0$) than radial layout ($Mdn =$

1.0); $W = 448.5$, $p = 0.27$, suggesting that participants had performed similarly in answering the questions.

(2) Eye tracking metrics

An independent two sample t-test to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear). The result indicated that **time to 1st fixation on the last of multiple targets** was greater for linear layout ($M = 16.46s$, $SD = 8.15$) than radial layout ($M = 11.32s$, $SD = 3.96$), $t(27) = 2.52$, $p = 0.018$. The eye tracking results support the result that, participants spend less time on radial layout than linear layout processing information from all the targets.

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **time after 1st fixation on the last of multiple targets**. There was no significant difference between linear layout ($M = 13.45s$, $SD = 6.90s$) and radial layout ($M = 11.31s$, $SD = 86.64$); $t(27) = 0.99$, $p = 0.33$, which indicates the processing time was similar for participants in terms of information on both radial and linear graph before they finally made their decisions.

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **total duration of fixations**. There was a significant difference in terms of total duration of fixations for linear layout ($M = 17.564$, $SD = 1.736s$) and radial layout ($M = 10.598s$, $SD = 0.889s$); $t(26) = 3.57$, $p = 0.001$. This suggests participants spent less time in total to process information on radial than linear layout.

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **mean fixation duration**. There was a significant difference between linear layout ($M = 236.2\text{ms}$, $SD = 31.3\text{ms}$) and radial layout ($M = 189.9\text{ms}$, $SD = 26.6\text{ms}$); $t(34) = 4.85$, $p < 0.001$. This suggests that on average, it takes less time for participants on each information processing point while fixating on radial layout than linear layout.

An independent two sample t-test to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear). The result indicated that **total duration of saccades** was no significant difference between linear layout ($M = 9.832\text{s}$, $SD = 2.571$) than radial layout ($M = 10.133\text{s}$, $SD = 4.398\text{s}$), $t(27) = -0.26$, $p = 0.80$. Generally, participants spend similar amount of time in the transitioning among all the information points on both linear and radial Infographics.

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **number of saccades**. There was a significant difference between linear layout ($M = 77.8$, $SD = 35.2$) and radial layout ($M = 56.4$, $SD = 12.8$); $t(32) = 2.27$, $p = 0.03$. This shows that participants make more movement on radial layout than on linear layout.

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **total amplitude of scanpath***. There was no significant difference between linear layout ($M = 379.3^\circ$, $SD = 198^\circ$) and radial layout ($M = 275.9^\circ$, $SD = 138.1^\circ$); $t(32) = 0.63$, $p > 0.53$. This tells that the total eye movement in arc degree was similar between

linear and radial Infographics; participants didn't spend more effort on either of the infographics in searching.

(Total amplitude of scanpath is the measure of scanpath length in arc degree)

An independent two sample t-test to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear). The result indicated that **mean saccade amplitude*** was no significant difference between linear layout ($M = 6.778^\circ$, $SD = 2.358^\circ$) than radial layout ($M = 6.93^\circ$, $SD = 4.59^\circ$), $t(27) = -0.87$, $p = 0.39$. This indicates that both infographics did not exhibit more useful visuals cues for the participants to find the correct pieces of information.

(Mean saccade amplitude is the measure of mean saccade length in arc degree)

An independent two sample t-test was conducted to compare Figure 3-1 Every Minute of the day (Radial) (left) and Figure 3-4 Every Minute of the day (Linear) with respect to **fixations/saccades ratio**. There was a significant difference between linear layout ($M = 1.747$, $SD = 0.580$) and radial layout ($M = 1.116$, $SD = 0.352$); $t(29) = 4.02$, $p < 0.001$. This shows participants on average spent significantly more time processing information than searching information on linear than radial infographics.

4.2.3 Trial 3 and Trail 4

Because of the nature of these two trials, only descriptive analysis of trial 3 and trial 4 were reported in Table 4-3.

Table 4-3 Descriptive analysis of trial 3 and trial 4

		<i>M,SD</i> <i>(Caffeine Intake)</i>	<i>M,SD</i> <i>(Where does the money go)</i>
Performance Metrics <i>(n = 40)</i>	Completion time (s)	64.42 ± 28.28	61.97 ± 26.37
	Accuracy score	<i>Mdn</i> = 2.5	<i>Mdn</i> = 3.5
Eye tracking metrics <i>(n = 38)</i>	Total duration of fixations (s)	29.61 ± 13.10	38.38 ± 18.97
	Mean fixation duration (ms)	193.81 ± 22.92	218.83 ± 31.00
	Total duration of saccades (s)	33.67 ± 17.35	23.24 ± 9.15
	Mean saccade duration(ms)	224.80 ± 63.80	138.88 ± 29.7
	Total amplitude of scanpath (°)	2024 ± 1429	1129 ± 578
	Mean saccade amplitude (°)	13.28 ± 6.65	6.89 ± 3.82
	Fixations/Saccades ratio	0.94 ± 0.31	1.66 ± 0.48

From the observation of the mean and standard deviation of metrics in Table 4-3, it appears that it takes much similar time for participants to read different infographics regardless of the contents (64.42s vs.61.97s); and participants seemed to have more long-distance eye movement in first infographics than the second (mean saccade amplitude).

4.3 Typical Scanpath Visualization by Infographics and Trial

Scanpaths visualization were created to study the search/scan pattern in this study. As there are 4 scanpaths per participants, it is not possible to list all the scanpaths here in this section. However, based on the average performance of participants in each trail, including average completion time, total duration of fixations and total duration of

saccades, 6 scanpaths were filtered out that best represent the average performance of participants per trial per picture.

(1) Value lookup on linear infographics



Figure 4-1 Value lookup - Linear

This participant started in the middle and quickly searched (the size of the circles of fixations are relatively small before the 28th fixation, meaning one spend only little time processing information) from very left and then to very right and then back to the very left and started more careful and longer reading (28th fixation and after) on each piece of the information until the target information was found.

(2) Value lookup on radial layout infographics

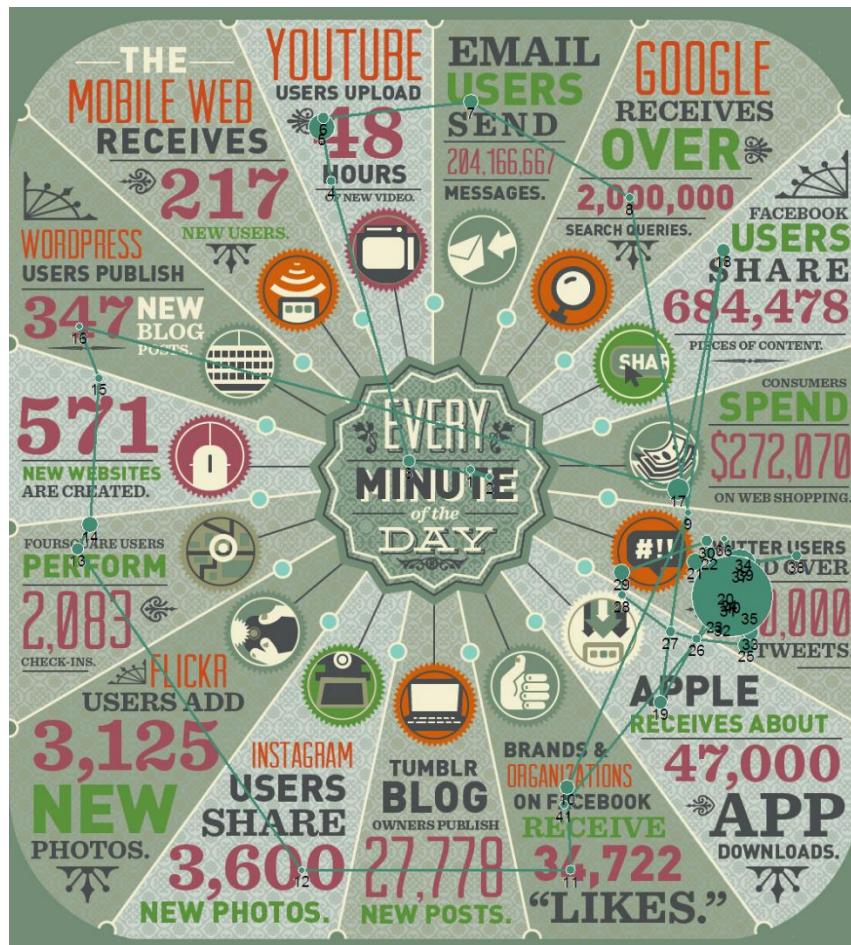


Figure 4-2 Value lookup - radial

The participant started at the center of the infographics and quickly (small size of fixation circles) went through the whole circle clockwise, skipping some of the chunks, and then jump back and forth (17th, 18th, 19th and 20th) until one finds the intended target.

(3) Value comparison on linear infographics

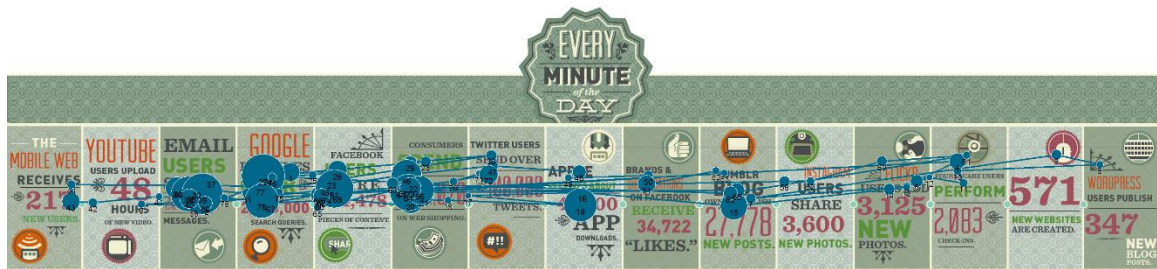


Figure 4-3 Value comparison - linear

There are too many overlapping fixations in this scanpath. However, through careful examination, phenomenon was observed that the participant very quickly go through most of the chunks in the first round, skipping some once in a while (starting at Google → YouTube → last one on the right); and then large fixations points showed at the 15th fixation point and comparisons in local area (chunks next to each other) begun (compare to global search and sensing in the first round), until the end of the task.

(4) Value comparison on radial infographics

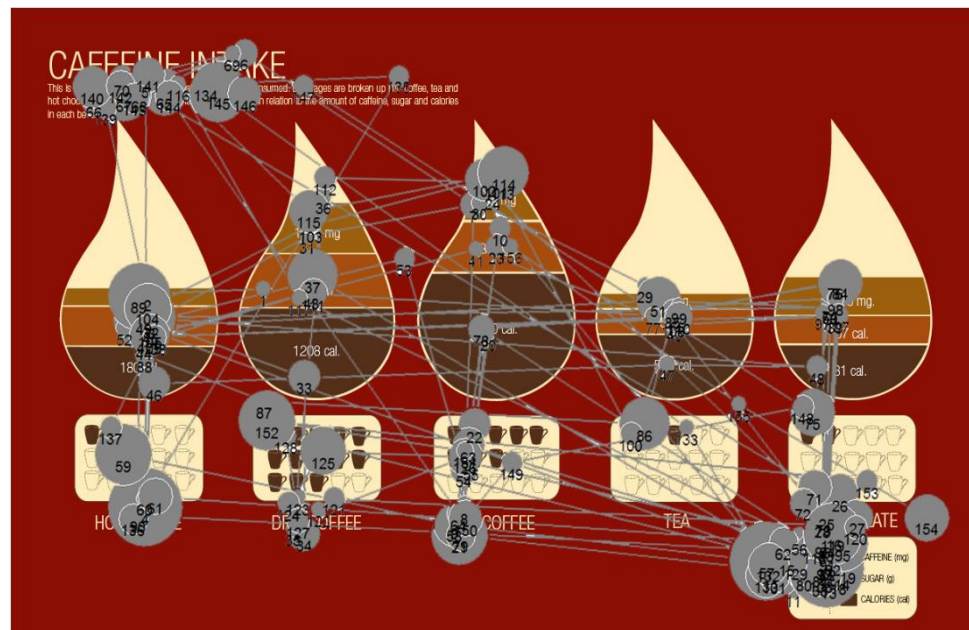


Figure 4-5 Information scanning - linear

There are too many fixations and saccades in this infographics that it is difficult to describe the scanpath verbally. One fact, though, can be observed is that the participant repeatedly referring the legend on the right lower corner.

(6) Information scanning on radial infographics

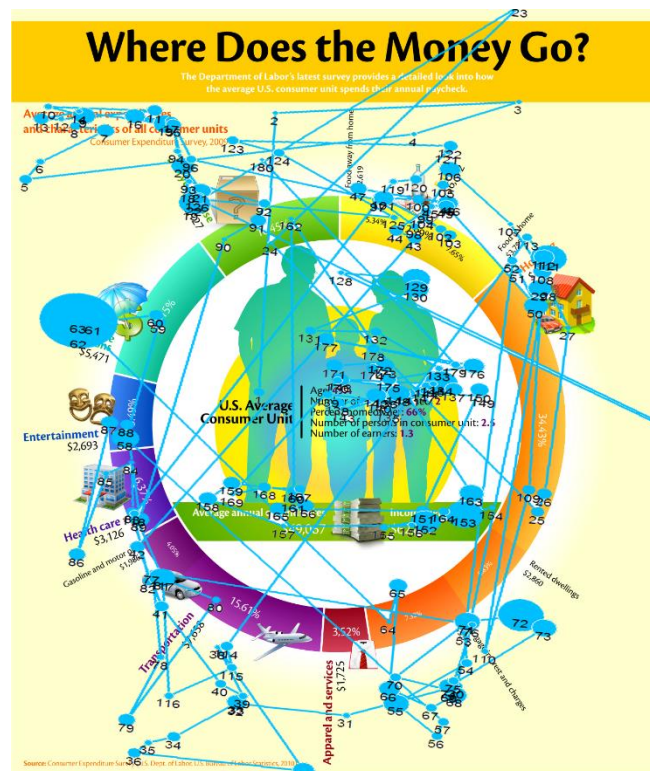


Figure 4-6 Information scanning - radial

There are too many fixations and saccades in this infographics that it is difficult to describe the scanpath verbally. One observation, though, is picture elements are rather eye-catching.

Chapter 5

DISCUSSION

The main finding of this research is that, on the first level of information perception and interpretation, the searching performance of infographics with linear layout is as effective as infographics of radial layout; on the second level of information where it requires comparison of values, the infographics with radial layout is more effective than infographics with linear layout.

In the comparison of values, participants who read the radial infographics outperformed participant who read the linear Infographics, with less time performing the task, faster time processing information, quicker time locating all the necessary pieces of information. Although the findings didn't indicate there is difference in the searching efficiency of the two layouts, the numbers of transition across information chunks were less in infographics with radial layout under the same amount of searching time, meaning more high quality information processing.

Goldberg & Helfman (2011) found that on the first level of information perception and interpretation (either one-dimension or two-dimension value lookup), radial graphs had worse performance than linear graphs. In this study, such result was not obtained: both were not significantly different than each other. The reasons might be that (1) the graphs Goldberg & Helfman used was more graphical integrated information; the layout of the graph itself is more dominant on the result; because the representation of the

forms are somewhat different, hence different results; (2) graphs used in Goldberg & Helfman's experiment were rather small in size (20 × 20 cm), and the infographics in this study could take up all the space along one-dimension (either height or width) of a 27-inches monitor; the size of the material itself might affect the performance (3) the nature of the task might be too easy, making performance of different participant indifferent among each other.

Also, Goldberg & Helfman (2010) didn't find difference between linear graphs and radial graphs on the second level of information perception and interpretation, however, this study obtained significant difference on radial layout over linear. The disagreement in the results might because (1) again, although the types of layout used were the same, the information arrangement was different, with Goldberg and Helfman's information more integrated in the graphs and ours more explicitly spread along the layout; (2) the easy task in Goldberg and Helfman's involved one decision making and hard task required more than 8 decisions and the use of short term memory to recall the result of each decision, while this study is neither the case; therefore participants' performance were different than in Goldberg and Helfman's experiment.

One of the limitations of this study is that the eye tracking device used recorded samples at a frequency of 30 Hz (recommended by the manual). This is a rather low frequency in eye tracking recording; some of the other devices could use frequency as high as 250 Hz or even more. The higher the frequency, the more information it records, hence the more accurate the results. Recording at 30Hz sampling frequency might introduce error up to 33ms. Due to the fast movement of eye (fastest can be achieved by

the body), there are lots of possibility within the 33ms error. Also, in the analysis part of this study, because of the sampling frequency, the minimum threshold of fixations which normally accepted as 100ms by the researchers, had to lower to 99ms, otherwise saccades captured by the algorithm would be as long as 500ms which is much higher the duration of saccades in the literature(30 – 80ms). However, even the time threshold dropped to 99ms, mean saccade duration of all the trials is still 140-150ms in this study, almost twice as much as the one in the literature. This indicates the more information about actual eye movement should have been captured. Through the examination and exploration of the accuracy of data recorded in this experiment, we found that while low frequency as 30 Hz might be sufficient for static task where eye movement is motivated by internal needs and influenced by personal reading behavior, it might not be able to obtain good results when it comes to dynamic tasks where human's information behavior is greatly affected by visual stimuli provided at the time. In addition, because of the low sampling frequency, the type of algorithm used to extract the eye tracking metrics was limited. Velocity-threshold was used at first, but it generates really bad results, such as fixations as long as 4 seconds.

Also, a head mounted eye tracker in glasses form was used in the study, however, there were a lot of participants who wear glassed already and had to wear another one which made them feel uncomfortable. The numbers of participants who wear glasses in each group was the factor that researchers failed to control, therefore might affect the results.

Descriptive analysis of scanpath was conducted through visual inspection on average performance by trial in the previous section, however, the study couldn't conclusively address the questions about the patterns in information searching and scanning due to the lack of efficient algorithms to detect the pattern along with the high complexity of the scanpaths. Further studies are needed to tackle this difficulty of identifying pattern on micro-behavior level separately on a simpler task.

Chapter 6

CONCLUSION

The types of infographics layout do have an impact of people's information behavior. The study found that in the design of infographics as a way of conveying information to the reader, when the context of the information involves values comparison, it is recommended to arrange the information in a radial way. The reason of doing so is that: during comparison, when eye moves from one point to another on a linear layout, it doesn't follow the direction of the shortest distance between the two points; the actual movement follows an arc route above or below the shortest distance, meaning more movement happens than it is supposed to be. However, this is rarely the case when reading information on a radial layout; the actual scanpath overlap with the shortest distance in most cases, therefore it's much easier for people to compare information on a radial layout.

Eye tracking is a great tool to study people's behavior on a micro level. It provides rich information of movement in millisecond time interval and supports and elaborate the macro behavior with much more evidence. It is always recommended to use a higher sampling frequency to catch the movement in order to get more accurate results from the eye tracking raw data.

Future studies could be design to further investigate people's searching behavior on the first level of information perception and interpretation using a more complex task

on infographics with more information context, not just explicitly list all the information. Also, because the first few largest numbers in the values comparison part of the study happened to lie next to each other, it would be interesting to see how people information behavior change when the pieces of information they need to compare is apart from each other. Lastly, there are already some preliminary observations about the pattern of information searching, which is people tend to start with global scanning first to get a first impression of the data, then narrow down to local sub-search; also they like to jump around to look for information. Therefore, further experiment needs to be specifically designed to study the behavior pattern in information searching and scanning with more careful experiment planning.

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APPENDIX A

Summary of Participants' Answers

Graph Type	Graph Number	Trial 1		Trial 2		Trial 3		Trial 4	
Linear	No.								
	1	1	Y	1	Y	1	Y	1	Y
	2	0	Y	0	Y	0	Y	0	Y
	3	1	Y	0	Y	1	Y	0	N
	4	1	Y	1	Y	0	Y	1	N
	5	1	Y	0	Y	0	Y	1	Y
	6	1	Y	0	Y	1	Y	1	Y
	7	1	Y	1	Y	1	N	0	Y
	8	1	Y	0	Y	1	Y	1	Y
	9	1	Y	1	Y	0	Y	1	Y
	10	1	Y	1	Y	1	Y	1	Y
	11	1	Y	0	N	1	Y	0	N
	12	1	Y	1	N	0	Y	1	Y
	13	1	Y	0	N	0	Y	1	N
	14	1	Y	1	N	1	Y	1	Y
	15	1	Y	0	N	0	Y	0	Y
	16	1	Y	0	N	1	Y	1	Y
	17	1	Y	0	N	0	Y	1	Y
	18	1	Y	0	N	1	Y	1	Y
	19	1	N	1	Y	0	N	0	N
	20	1	Y	1	N	1	Y	1	Y
Graph Type	Graph Number	1-1		1-2		2		3	
Radial	No.								
	1	1	Y	1	Y	1	Y	0	Y
	2	1	Y	0	N	0	N	0	Y
	3	1	Y	0	N	1	Y	0	Y
	4	1	Y	0	N	0	Y	1	N
	5	1	Y	0	Y	1	N	1	Y
	6	1	Y	1	Y	0	N	0	N
	7	1	Y	1	Y	0	Y	1	Y
	8	1	Y	0	N	1	Y	1	Y
	9	1	Y	0	Y	1	Y	1	N

10	1	Y	1	Y	0	N	1	Y
11	1	Y	0	N	1	N	1	Y
12	1	Y	0	N	0	N	1	Y
13	1	Y	0	N	0	Y	1	Y
14	1	Y	1	Y	1	Y	1	Y
15	1	Y	1	N	0	N	1	Y
16	0	N	1	N	0	N	1	N
17	1	Y	0	N	1	Y	1	Y
18	1	Y	1	Y	0	N	0	Y
19	1	Y	0	N	0	N	1	Y
20	1	Y	0	Y	0	Y	1	N

APPENDIX B

Results of Statistics Test

Trial 1

	Norm	Transform	Equal Var	Test	<i>M,SD</i>	<i>t/W/Z</i>	<i>p</i>
Completion time		√	√	Two sample t-test	<i>M1</i> = 21.499 <i>SD1</i> = 15.499 <i>M2</i> = 17.054 <i>SD2</i> = 8.210	t(36) = -0.93	0.36
Accuracy score	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Time to 1st fixation on intended target		√	√	Two sample t-test	<i>M1</i> = 15.82 <i>SD1</i> = 13.65 <i>M2</i> = 11.74 <i>SD2</i> = 7.75	t(36) = 1.18	0.24

Time to 1st fixation on the last of multiple targets	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Time after 1st fixation on intended target		√	√	Two sample t-test	$M1 = 5.966$ $SD1 = 2.758$ $M2 = 5.839$ $SD2 = 2.981$	$t(36) = 0.15$	0.88
Total duration of fixations		√	√	Two sample t-test	$M1 = 13.721$ $SD1 = 13.014$ $M2 = 9.018$ $SD2 = 4.094$	$t(34)=1.57$	0.13
Mean fixation duration	√		√	Two sample t-test	$M1 = 213.1$ $SD1 = 40.8$ $M2 = 190.3$ $SD2 = 22.7$	$t(33) = 2.01$	0.052
Total duration of saccades	√		√	Two sample t-test	$M1 = 7.778$ $SD1 = 0.705$ $M2 = 6.604$ $SD2 = 0.483$	$t(32) = 1.37$	0.18
Mean saccade duration	√		√	Two sample t-test	$M1 = 147.9$ $SD1 = 45.4$ $M2 = 154.0$ $SD2 = 53.2$	$t(27) = -0.41$	0.683
Total amplitude of scanpath		√	√	Two sample t-test	$M1 = 379.3$ $SD1 = 198.0$ $M2 = 275.9$ $SD2 = 138.1$	$t(34) = 1.66$	0.11
Mean saccade amplitude		√	√	Two sample t-test	$M1 = 6.778$ $SD1 = 2.358$ $M2 = 6.93$ $SD2 = 4.59$	$t(25) = 0.65$	0.53
Fixations/Saccades ratio		√	√	Two sample t-test	$M1 = 1.663$ $SD1 = 0.797$ $M2 = 1.414$ $SD2 = 0.577$	$t(32) = 1.31$	0.20

Trial 2

	Norm	Transform	Equal Var	Test	<i>M,SD</i>	<i>t/W/Z</i>	<i>p</i>
Completion time	√		√	Two sample t-test	<i>M1</i> = 29.439 <i>SD1</i> = 13.127 <i>M2</i> = 21.947 <i>SD2</i> = 8.936	t(33) = 2.09	0.044
Accuracy score	N/A		√	Mann Whitney U test	<i>Mdn1</i> = 2.0 <i>Mdn2</i> = 1.0	W = 448.5	0.27
Time to 1st fixation on intended target	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Time to 1st fixation on the last of multiple targets	√		√	Two sample t-test	<i>M1</i> = 16.46 <i>SD1</i> = 8.15 <i>M2</i> = 11.32 <i>SD2</i> = 3.96	t(27) = 2.52	0.018

Time after 1st fixation on intended target	√		√	Two sample t-test	$M1 = 13.45$ $M2 = 11.31$	$SD1 = 6.90$ $SD2 = 6.64$	$t(36) = 0.99$	0.33
Total duration of fixations	√		√	Two sample t-test	$M1 = 17.564$ $M2 = 10.598$	$SD1 = 1.736$ $SD2 = 0.889$	$t(26) = 3.57$	0.001
Mean fixation duration	√		√	Two sample t-test	$M1 = 236.2$ $M2 = 189.9$	$SD1 = 31.3$ $SD2 = 26.6$	$t(34) = 4.85$	<0.001
Total duration of saccades	√		√	Two sample t-test	$M1 = 9.823$ $M2 = 10.133$	$SD1 = 2.571$ $SD2 = 4.398$	$t(27) = -0.26$	0.80
Number of saccades	√		√	Two sample t-test	$M1 = 77.8$ $M2 = 56.4$	$SD1 = 35.2$ $SD2 = 21.8$	$t(32) = 2.27$	0.030
Total amplitude of scanpath		√	√	Two sample t-test	$M1 = 379.3$ $M2 = 275.9$	$SD1 = 198$ $SD2 = 138.1$	$t(32) = 0.63$	0.53
Mean saccade amplitude		√	√	Two sample t-test	$M1 = 7.07$ $M2 = 8.41$	$SD1 = 2.59$ $SD2 = 4.65$	$t(34) = -0.87$	0.39
Fixations/Saccades ratio	√		√	Two sample t-test	$M1 = 1.747$ $M2 = 1.116$	$SD1 = 0.580$ $SD2 = 0.352$	$t(29) = 4.02$	<0.001